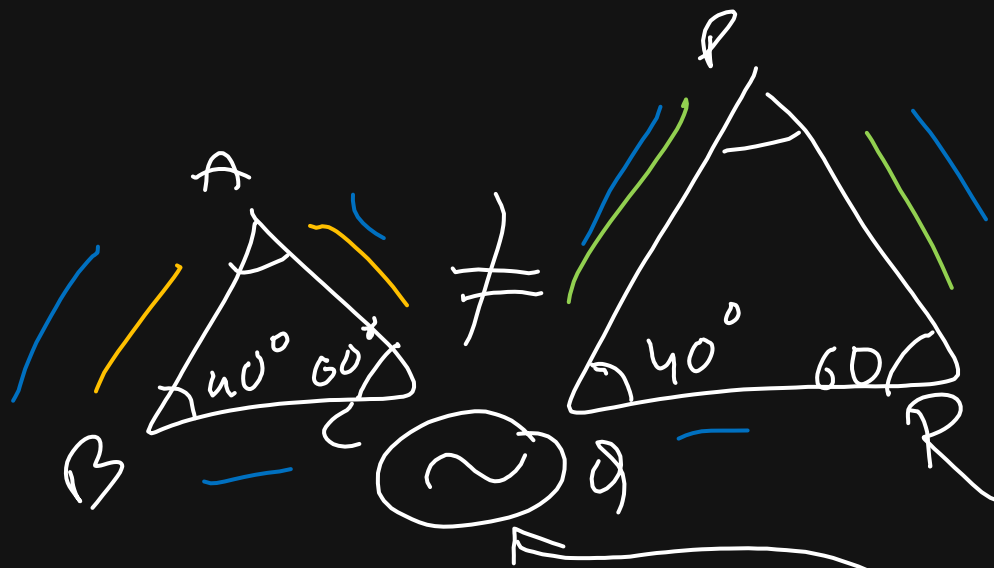




$$\frac{AB}{AC} = \frac{PQ}{QR}$$

$$\frac{A(ABC)}{A(PQR)} =$$

Area of $\triangle ABC$

$$\frac{A(ABC)}{A(PQR)} = \frac{BC}{QR} = \frac{AC}{PR}$$

$$\sqrt{x} = x$$

$$2\sqrt{x} = (x-2)$$

$$3\sqrt{x} = (x+3)$$

$$(x-2)(x+3) = 84$$

$$x(x+3) - 2(x+3) = 84$$

$$x^2 + 3x - 2x - 6 = 84$$

$$x^2 + x - 6 \quad \leftarrow = +84$$

$$x^2 + x - 6 - 84 = 0$$

$$x^2 + x - 90 = 0$$

and
/

$$x = \boxed{0}$$

